

# Model Predictive Control

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github.com/silvasta/summary-mpc



## Contents

|      |                                              |   |
|------|----------------------------------------------|---|
| 1    | Nominal MPC                                  | 1 |
| 2    | Practical MPC                                | 1 |
| 2.1  | Steady-state Target Problem                  | 1 |
| 2.2  | Offset-free Reference Tracking               | 1 |
| 2.3  | Soft Constraints                             | 1 |
| 3    | Robust MPC                                   | 2 |
| 3.1  | Robust Constraint-Tightening MPC             | 2 |
| 3.2  | Robust Tube MPC                              | 2 |
| 4    | Stochastic MPC                               | 2 |
| 5    | Nonlinear MPC                                | 2 |
| 5.1  | Optimization                                 | 2 |
| 5.2  | Robust NMPC                                  | 2 |
| 5.3  | Stochastic NMPC                              | 3 |
| 6    | Parameter Uncertainty in MPC                 | 3 |
| 7    | Safety Filter                                | 3 |
| 8    | Learning Based MPC                           | 3 |
| 9    | Invariance (and more)                        | 3 |
| 9.1  | Control Invariance                           | 3 |
| 9.2  | Robust Invariance                            | 3 |
| 9.3  | Computing Invariant Sets and Pre-sets        | 3 |
| 10   | Optimization                                 | 3 |
| 10.1 | Convex Sets                                  | 3 |
| 10.2 | Convex Functions                             | 4 |
| 10.3 | Optimality Conditions                        | 4 |
| 10.4 | Convex Optimization Problems                 | 4 |
| 11   | Introduction to Systems and Controls         | 4 |
| 11.1 | Linearization                                | 4 |
| 11.2 | Discretization                               | 4 |
| 11.3 | Analysis of Discrete-Time LTI Systems        | 4 |
| 12   | Optimal Control                              | 4 |
| 12.1 | Unconstrained Finite Horizon Control Problem | 4 |
| 12.2 | Infinite Horizon Control Problem             | 4 |

## 12.3 Constrained Finite Time Optimal Control Problem

### Requirements and Steps to MPC

- 1 **Model of the System** dynamics to state space
- 2 **State Estimator** track trajectory and disturbance
- 3 **Optimal Control Problem** define control strategy
- 4 **Optimization problem** mathematical formulation
- 5 **Get Optimal Control Sequence** solve op in time
- 6 **Verify Closed-Loop Performance** iterative tests

## 1 Nominal MPC

### MPC Mathematical Formulation

$$\operatorname{argmin}_U \sum_{i=0}^{N-1} I(x_i, u_i) + I_f(x_N) \quad (1)$$

**Constraints**  $x_0 = x(k)$   
 $x_{i+1} = f(x_i, u_i)$  linear case :  $Ax_i + Bu_i$   
 $x_i \in \mathcal{X}$   
 $u_i \in \mathcal{U}$   
 $x_N \in \mathcal{X}_f$   
 $I_f(\cdot)$  and  $\mathcal{X}_f$  are chosen to mimic an infinite horizon.

What can go wrong with *standard* MPC?

- No feasibility guarantee, the problem may not have a solution
- No stability guarantee, trajectories may not converge to origin

### Stability of MPC - Main Result

#### Assumptions

- 1 Stage cost is strictly positive and only zero at the origin
- 2 Terminal set is **invariant** under local control law  $\kappa_f(x_i)$ :

$$x_{i+1} = Ax_i + B\kappa_f(x_i) \in \mathcal{X}_f \quad \forall x_i \in \mathcal{X}_f$$

All state and input **constraints are satisfied** in  $\mathcal{X}_f$ :

$$\mathcal{X}_f \in X, \kappa_f(x_i) \in U \quad \forall x_i \in \mathcal{X}_f$$

- 3 Terminal cost is a continuous **Lyapunov function** s.t.

$$l_f(x_{i+1}) - l_f(x_i) \leq -l(x_i, \kappa_f(x_i)) \quad \forall x_i \in \mathcal{X}_f$$

**Theorem 1.** Under the previous assumptions, the closed-loop system under the MPC control law  $u_0^*(x)$  is asymptotically stable and the set  $\mathcal{X}_f$  is positive invariant for

$$x(k+1) = Ax(k) + Bu_0^*(x(k))$$

**Same thing for non-linear MPC!** (f/g(x,u) instead of A/B)

**Finite-horizon MPC may not satisfy constraints for all time!**

**Finite-horizon MPC may not be stable!**

- An infinite-horizon provides stability and invariance.
- Infinite-horizon *faked* by forcing final state into an invariant

5 set for which there exists invariance-including controller, whose infinite-horizon cost can be expressed in closed-form.

- Extends to non-linear systems, but compute sets is difficult!

## 2 Practical MPC

### 2.1 Steady-state Target Problem

- Reference is achieved by the target state  $x_s$  if  $z_s = Hx_s = r$

- Target state should be a steady-state, i.e.  $x_s = Ax_s + Bu_s$

$$\begin{aligned} x_s &= Ax_s + Bu_s \\ z_s &= Hx_s = r \end{aligned} \iff \begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

‡ solution  $\rightarrow \min(Hx_s - r)^\top Q_s(Hx_s - r)$  (closest  $x$  to  $r$ )

If  $\exists$  multiple feasible  $u_s \rightarrow$  compute cheapest:  $\min u_s^\top R_s u_s$

$$\min_U |z_N - Hx_s|_{P_z}^2 + \sum_{i=1}^{N-1} |z_i - Hx_s|_{Q_z}^2 + |u_i - u_s|_R^2$$

### 2.2 Offset-free Reference Tracking

$$\begin{aligned} \Delta x_{k+1} &= x_{k+1} - x_s \\ &= A\Delta x_k + Bu_k - (Ax_s + Bu_s) \\ &= A\Delta x_k + B\Delta u_k \end{aligned}$$

$$\begin{aligned} G_x x &\leq h_x \Rightarrow G_x \Delta x \leq h_x - G_x x_s \\ G_u u &\leq h_u \Rightarrow G_u \Delta u \leq h_u - G_u u_s \end{aligned}$$

#### Convergence

Assume feasible target with  $x_s \in \mathcal{X}, u_s \in \mathcal{U}$ , choose terminal weight  $V_f(x)$  and constraint  $\mathcal{X}_f$  as in regulation case satisfying

$$V_f(x(k+1)) - V_f(x(k)) \leq -I(x(k), Kx(k))$$

and  $(A + BK)x \in \mathcal{X} \quad \forall x \in \mathcal{X}_f$  for both

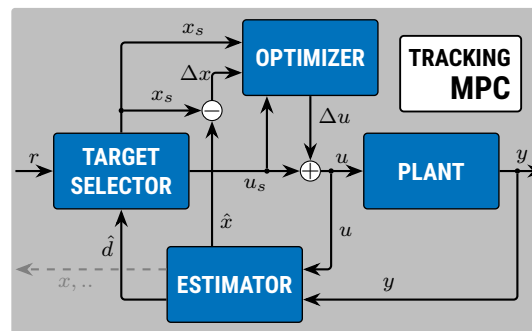
If in addition the target reference  $x_s, u_s$  is such that

$$x_s \oplus \mathcal{X}_f \subseteq \mathcal{X}, K\Delta x + u_s \in \mathcal{U}, \quad \forall \Delta x \in \mathcal{X}_f$$

then the closed loop system converges to the target reference.

**Proof.** Invariance under local control law is inherited from regulation case. Constraint satisfaction is provided by extra conditions and convergence comes from the asymptotic stability of the regulation problem:  $\Delta x(k) \rightarrow 0$  for  $k \rightarrow \infty$  □

**Terminal set** use  $\mathcal{X}_f^{\text{scaled}} = \alpha \mathcal{X}_f$  (s.t. constraints satisfied)



## Disturbance Cancellation

**Approach** Model the disturbance, use the measurements and model to estimate the state and disturbance and find control inputs that use the disturbance estimate to remove offset.

### Augmented Model

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + B_d d_k \\ y_k &= Cx_k + C_d d_k \end{aligned}$$

**Constant disturbance**  $d_{k+1} = d_k$

Observable iff  $\begin{bmatrix} A-I & B_d \\ C & C_d \end{bmatrix}$  has full rank (assuming  $n_x = n_d$ )

### Observer For Augmented Model

$$\begin{bmatrix} \hat{x}_{k+1} \\ \hat{d}_{k+1} \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}_k \\ \hat{d}_k \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k + \begin{bmatrix} L_x \\ L_d \end{bmatrix} (Cx_k + C_d \hat{d}_k - y_k)$$

**Error Dynamics**  $\Rightarrow$  choose  $L$  s.t error dynamics converge to 0

$$\begin{bmatrix} x_{k+1} - \hat{x}_{k+1} \\ d_{k+1} - \hat{d}_{k+1} \end{bmatrix} = \left( \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} C & C_d \end{bmatrix} \right) \begin{bmatrix} x_k - \hat{x}_k \\ d_k - \hat{d}_k \end{bmatrix}$$

**Lemma 1.** Steady-state of an asym. stable observer satisfies:

$$\begin{bmatrix} A-I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ y_\infty - C_d \hat{d}_\infty \end{bmatrix} \quad (\text{for } n_y = n_d)$$

$\Rightarrow$  Observer output  $C\hat{x}_\infty + C_d \hat{d}_\infty$  tracks  $y_\infty$  without offset

### Reference Tracking with Disturbance Cancellation

**Goal** Track constant reference:  $Hy(k) = z(k) \rightarrow r, k \rightarrow \infty$

$$\begin{aligned} x_s &= Ax_s + Bu_s + B_d \hat{d}_\infty \\ z_s &= H(Cx_s + C_d \hat{d}_\infty) = r \end{aligned}$$

$$\begin{bmatrix} A-I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r - HC_d \hat{d} \end{bmatrix}$$

### Offset-free Tracking - Main Result

**Theorem 2.** Assuming RHC recursively feasible,  $n_d = n_y$ , unconstrained for  $k \geq j$ , and the closed loop system

$$x(k+1) = Ax(k) + B\kappa(\cdot) + B_d d \quad \text{with } (\cdot) = (\hat{x}, \hat{d}, r)$$

$$\hat{x}(k+1) = (A + L_x C)\hat{x}(k) + (B_d + L_x C_d)\hat{d}(k)$$

$$+ B\kappa(\cdot) - L_x y(k)$$

$$\hat{d}(k+1) = L_d C\hat{x}(k) + (I + L_d C_d)\hat{d}(k) - L_d y(k)$$

converges, then  $z(k) = Hy(k) \rightarrow r$  as  $k \rightarrow \infty$

## 2.3 Soft Constraints

**Input constraints** are dictated by physical constraints on the actuators and are **usually hard**

**State/output constraints** arise from practical restrictions on the allowed operating range and are **rarely hard**

**Hard state/output constraints** always lead to **complications in the controller implementation**

## Soft Constrained MPC

$$\min_{u \in \mathcal{P}S} \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i + l_\epsilon(\epsilon_i) + x_N^\top P x_i + l_\epsilon(\epsilon_N) \quad (2)$$

**Quadratic penalty**  $l_\epsilon(\epsilon_i) = \epsilon_i^\top S \epsilon_i$  (e.g  $S = Q$ )  
**+ linear norm penalty**  $l_\epsilon(\epsilon_i) = v |\epsilon_i|_1 / \infty$

**Constraints**

$$\begin{aligned} x_{i+1} &= A x_i + B u_i \\ H x_i &\leq k x + \epsilon_i \\ H u_i &\leq k u \\ \epsilon_i &\geq 0 \text{ slack variable} \end{aligned}$$

|                 |                    |                 |                                                  |
|-----------------|--------------------|-----------------|--------------------------------------------------|
| <b>Original</b> | $\min_z f(z)$      | <b>Softened</b> | $\min_{z, \epsilon} f(z) + l_\epsilon(\epsilon)$ |
|                 | s.t. $g(z) \leq 0$ |                 | s.t. $g(z) \leq \epsilon$<br>$\epsilon \geq 0$   |

**Requirement on  $l_\epsilon(\epsilon)$**  If the original problem has a feasible solution  $z^*$ , then the softened problem should have the same solution  $z^*$ , and  $\epsilon = 0$ .

**Theorem 3** (Exact Penalty Funtcion).  $l_\epsilon(\epsilon) = v \cdot \epsilon$  satisfies requirement for any  $v > \lambda^* \geq 0$ , where  $\lambda^*$  is optimal Lagrange multiplier for original problem

## 3 Robust MPC

### Uncertain System

$$x(k+1) = g(x(k), u(k), w(k); \theta)$$

### Min-Max Robust Formulation

$$\begin{aligned} \min_U \max_W \sum_{i=1}^N \ell(x_{i|k}, u_{i|k}) \\ \text{s.t. } x_{i+1|k} &= A x_{i|k} + B u_{i|k} + w_{i|k}, \\ \forall w_{i|k} \in \mathcal{W} \quad x_{i|k} &\in \mathcal{X} \quad u_{i|k} \in \mathcal{U} \end{aligned}$$

### Robust Constraint Satisfaction

**Idea** Compute a set of tighter constraints such that if the nominal system meets these constraints, then the uncertain system will too. We then do MPC on the nominal system.

**Goal** Ensure constraints satisfied for the MPC sequence.

**Disturbance reachable set**  $\mathcal{F}_i = \bigoplus_{j=0}^{i-1} A^j \mathcal{W}$

## Robust Open-Loop MPC

$$\begin{aligned} \min_U \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N) \\ \text{s.t. } x_{i+1} &= A x_i + B u_i \\ x_0 &= x(k) \\ x_i &\in \mathcal{X} \ominus \mathcal{F}_i \\ u_i &\in \mathcal{U} \\ x_N &\in \mathcal{X}_f \ominus \mathcal{F}_N \end{aligned}$$

### Closed Loop Robust MPC

**Idea** Separate the available control authority into two parts:

$$z(k+1) = A z(k) + B v(k)$$

steers noise-free *nominal* system to origin

$$u_i = K(x_i - z_i) + v_i$$

compensates for deviations, i.e. a *tracking* controller, to keep the real trajectory close to the nominal system.  $\Rightarrow$  We fix the

linear feedback controller  $K$  offline and optimize over the nominal inputs  $\{v_0, \dots, v_{N-1}\}$  and nominal trajectory  $\{z_0, \dots, z_N\}$ , which results in a convex problem.

$$e_{i+1} = x_{i+1} - z_{i+1} = (A + BK)e_i + w_i$$

### 3.1 Robust Constraint-Tightening MPC

## Robust Constraint-Tightening MPC

$$\begin{aligned} \min_{Z, V} \sum_{i=0}^{N-1} l(z_i, v_i) + l_f(z_N) \quad & z_0 = x(k) \\ \text{subj. to } z_{i+1} &= A z_i + B v_i \quad & z_i &\in \mathcal{X} \ominus \mathcal{F}_i \\ & & u_i &\in \mathcal{U} \ominus K \mathcal{F}_i \\ & & z_N &\in \mathcal{X}_f \ominus \mathcal{F}_N \end{aligned}$$

$$\mathcal{F}_0 := 0 \quad \mathcal{F}_i := \mathcal{W} \oplus A_K \mathcal{W} \oplus \dots \oplus A_K^{i-1} \mathcal{W}$$

**Control Law**  $u(k) = v_0^* + K(x(k) - z_0) = v_0^*$

### 3.2 Robust Tube MPC

**Idea** Ignore noise and plan the nominal trajectory, bound maximum error at any time with RPI set  $\mathcal{E} : \epsilon_i \in \mathcal{E} \epsilon_{i+1} \in \mathcal{E}$

Ideally  $\mathcal{E}$  is selected as the minimum RPI set  $\mathcal{F}_\infty$

We know that the real trajectory stays ‘nearby’ the nominal one  $x_i \in z_i \oplus \mathcal{E}$  because we plan to apply the controller  $u_i = K(x_i - z_i) + v_i$  in the future.

(we won’t actually do this, but it’s a valid sub-optimal plan)

We must ensure that all possible state trajectories satisfy the constraints This is now equivalent to ensuring that  $x_i \in z_i \oplus \mathcal{E}$  What do we need to make this work?

**Compute the set  $\mathcal{E}$  that the error will remain inside**

Previously we wanted the **maximum robust invariant set**, or the largest set in which our terminal control law works.

We now want the **minimum robust invariant set**, or the smallest set that the state will remain inside despite the noise.

Modify constraints on nominal trajectory  $\{z_i\}$

$$x_i \in z_i \oplus \mathcal{E} = \{z_i + e | e \in \mathcal{E}\}$$

## Tube MPC

$$\begin{aligned} \text{Cost function } J(Z, V) &:= \sum_{i=0}^{N-1} l(z_i, v_i) + l_f(z_N) \\ \text{Feasible set } \mathcal{Z}(x_0) &:= \left\{ \begin{aligned} z_{i+1} &= A z_i + B v_i \\ z_i &\in \mathcal{X} \ominus \mathcal{E} \\ v_i &\in \mathcal{U} \ominus K \mathcal{E} \\ z_N &\in \mathcal{X}_f \\ x_0 &\in z_0 \oplus \mathcal{E} \end{aligned} \right. \end{aligned}$$

**Optimization**  $(V^*(x_0), Z^*(x_0)) = \operatorname{argmin}_{V, Z} \{J(Z, V) | (Z, V) \in \mathcal{Z}(x_0)\}$

**Control law**  $\mu_{\text{tube}}(x) := K(x - z_0^*(x)) + v_0^*(x)$

**Theorem 4** (Robust Invariance of Tube MPC). The set  $\mathcal{Z} := \{x | \mathcal{Z}(x) \neq \emptyset\}$  is a robust invariant set of the system  $x(k+1) = A x(k) + B \mu_{\text{tube}}(x(k)) + w(k)$  subject to the constraints  $x, u \in \mathcal{X} \times \mathcal{U}$ .

**Theorem 5** (Robust Stability of Tube MPC). The state  $x(k)$  of

the system  $x(k+1) = A x(k) + B \mu_{\text{tube}}(x(k)) + w(k)$  converges to the limit of the set  $\mathcal{E}$ .

### Tube MPC - Quick Summary

To implement tube MPC:

– **Offline** –

1. Stabilizing controller  $K$  so that  $A + BK$  is (Schur) stable
2. Compute the minimal robust invariant set  $\mathcal{E} = \mathcal{F}_\infty$  for the system  $x(k+1) = (A + BK)x(k) + w(k)$ ,  $w \in \mathcal{W}$
3. Compute tightened constraints  $\bar{\mathcal{X}} := \mathcal{X} \ominus \mathcal{E}$ ,  $\bar{\mathcal{U}} := \mathcal{U} \ominus K \mathcal{E}$
4. Choose terminal weight function  $l_f$  and constraint  $\mathcal{X}_f$  satisfying assumptions\*

– **Online** –

1. Measure / estimate state  $x$
2. Solve optimization problem for  $(V^*(x_0), Z^*(x_0))$
3. Set the input to  $u = K(x - z_0^*(x)) + v_0^*(x)$

## 4 Stochastic MPC

Stochastic MPC extends deterministic MPC to handle uncertainties modeled as random variables, such as additive noise or parameter variations. A core feature is replacing hard constraints with chance constraints to account for uncertainty.

$$\mathbb{P}(x_k \in \mathcal{X}) \geq 1 - \epsilon, \quad \mathbb{P}(u_k \in \mathcal{U}) \geq 1 - \delta,$$

This softens constraints, improving feasibility over robust MPC.

**Mean Dynamics**  $\bar{x}_{k+1} = A \bar{x}_k + B \bar{u}_k$  (Deterministic)

**Variance Dynamics**  $\Sigma_{k+1} = (A + BK)\Sigma_k(A + BK)^\top + \Sigma_w$

The variance  $\Sigma_k$  can be precomputed offline as it does not depend on the actual state measurements.

## Stochastic MPC

$$J = \mathbb{E} \left[ \|x_N\|_P^2 + \sum_{k=0}^{N-1} \|x_k\|_Q^2 + \|u_k\|_R^2 \right]$$

Using the property  $\mathbb{E}[x^\top Q x] = \bar{x}^\top Q \bar{x} + \operatorname{tr}(Q \operatorname{var}(x))$ , the cost decomposes into:

**Nominal Cost** A standard quadratic cost on the mean trajectories  $(\bar{x}, \bar{u})$ .

**Constant Offset** A term involving  $\operatorname{tr}(Q \Sigma_k)$  which depends only on the noise statistics and the gain  $K$ .

**Conclusion** For a fixed  $K$ , optimizing the expected cost is equivalent to optimizing the nominal cost of the mean system.

**System dynamics**  $x_{i+1} = A x_i + B u_i + w_i$  (probabilistic)  
**Chance constraints**  $\mathbb{P}(x_i \in \mathcal{X}, u_i \in \mathcal{U}) \geq 1 - \epsilon$   
**Initial condition**  $x_0 = x_k$   
**Terminal set**  $x_N \in \mathcal{X}_f$  (probabilistically)

**Tube-like Feedback Policy** (to control spread of trajectories)

$$u(k) = \bar{u}_k + K(x(k) - \bar{x}_k)$$

For tractability, approximations like affine feedback policies  $u_i = K x_i + v_i$  (where  $v_i$  is optimized) are used to close the loop in predictions.

For general distributions or polytopic sets, we use

**Probabilistic Reachable Set**  $\mathcal{F}_k^p$  such that  $\Pr(e(k) \in \mathcal{F}_k^p) \geq p$

The constraint is satisfied if:  $\bar{x}_k \in \mathcal{X} \ominus \mathcal{F}_k^p$

### Feasibility

Large (unbounded) disturbances can drive the system into regions where the optimization problem becomes infeasible at the next time step.

### Asymptotic Average Performance bound

Stability definition  $\lim_{T \rightarrow \infty} \frac{1}{T} \sum \mathbb{E}[l(x, u)] \leq \operatorname{tr}(P \Sigma_w)$

### Recovery Initialization

If the measured state  $x(k)$  is infeasible for the MPC, the solver is initialized with the **predicted mean** from the previous step ( $\bar{x}_{1|k-1}$ ) instead of the actual measurement. This ensures recursive feasibility but sacrifices performance.

### Indirect Feedback SMPC

A more advanced formulation where the nominal state  $z(k)$  evolves independently of the measurement  $x(k)$  in the constraints, but  $x(k)$  enters the **cost function**.

- **Pro** Guarantees recursive feasibility and satisfaction of closed-loop chance constraints.

- **Con** The feedback is “weaker” than direct state-feedback MPC.

## 5 Nonlinear MPC

### 5.1 Optimization

Because NMPC requires solving NLPs in real-time, algorithmic efficiency is critical.

**Sequential Quadratic Programming (SQP)** The standard method for NMPC. It solves a sequence of Quadratic Programs (QP) that approximate the NLP locally using a quadratic cost and linearized constraints.

### Sequential vs. Simultaneous Methods

**Sequential** (Single-shooting) Only control inputs are decision variables, states are eliminated via simulation. Results in small, dense QPs.

**Simultaneous** (Multiple-shooting) Both states and inputs are decision variables. Results in large, sparse QPs. (Preferred in practice for better convergence and handling of unstable dynamics)

**Real-Time Iteration (RTI)** To meet strict timing constraints, RTI performs only one SQP iteration per sampling time, using warm-starts from the previous step.

### 5.2 Robust NMPC

In linear systems, the error dynamics  $e = x - z$  are independent of the nominal state. In nonlinear systems, the **error dynamics** are **state-dependent**, making robust design much harder.

### Incremental Stability

**Incremental Lyapunov function**  $V_\delta(x, z)$  such that:

$$V_\delta(f(x, u), f(z, v)) \leq \rho V_\delta(x, z)$$

This characterizes how “close” the real trajectory  $x$  stays to the nominal trajectory  $z$ .

**Nonlinear Homothetic Tubes** The tube is defined as the level set of the incremental Lyapunov function:  $\mathbb{X}_i = \{x | V_\delta(x, z_i) \leq \delta_i\}$ . The tube size  $\delta_i$  is propagated dynamically:

$$\delta_{i+1} = \bar{\rho} \delta_i + \bar{w}(z_i, v_i)$$

where  $\bar{w}$  accounts for the worst-case model mismatch/disturbance.

**Constraint Tightening** Nominal constraints are tightened based on the current tube size  $\delta_i$ , ensuring that if the nominal state

is in the tightened set, the real state is guaranteed to satisfy original constraints.

### 5.3 Stochastic NMPC

SNMPC aims for **PRS** rather than worst-case tubes.

The constraints are tightened using the PRS. The cost function is typically the **expected cost** conditioned on the current state.

### 6 Parameter Uncertainty in MPC

### 7 Safety Filter

### 8 Learning Based MPC

### 9 Invariance (and more)

**Definition 1** (Positively Invariant Set  $\mathcal{O}$ ). For an autonomous or closed-loop system, the set  $\mathcal{O}$  is positively invariant if:

$$x(k) \in \mathcal{O} \Rightarrow x(k+1) \in \mathcal{O}, \quad \forall k \in \{0, 1, \dots\}$$

**Definition 2** (Maximal Positively Invariant Set  $\mathcal{O}_\infty$ ). A set that contains all  $\mathcal{O}$  is the maximal positively invariant set  $\mathcal{O}_\infty \subset \mathcal{X}$

**Definition 3** (Pre-Sets). The set of states that in the dynamic system  $x(k+1) = g(x(k))$  in one time step evolves into the target set  $S$  is the **pre-set** of  $S \Rightarrow \text{pre}(S) := \{x \mid g(x) \in S\}$

**Lemma 2.** **Invariant Sets from Lyapunov Functions**

If  $V: \mathbb{R}^n \rightarrow \mathbb{R}$  is a Lyapunov function for  $x(k+1) = g(x(k))$ , then  $Y := \{x \mid V(x) \leq \alpha\}$  is an invariant set for all  $\alpha \geq 0$

*Proof.* Lyapunov property  $V(g(x)) - V(x) < 0$  implies that once  $V(x(k)) \leq \alpha$ ,  $V(x(j)) < \alpha, \forall j \geq k \rightarrow$  Invariance  $\square$

**Example System**  $x(k+1) = Ax(k)$ ,  $A^\top PA - P \prec 0 \prec P$  and resulting Lyapunov function  $V(x(k)) = x(k)^\top Px(k)$

**Goal** Find the largest  $\alpha$  s.t the invarinat set  $Y_\alpha \subset \mathcal{X}$

$$Y_\alpha := \{x \mid x^\top Px \leq \alpha\} \subset \mathcal{X} := \{x \mid Fx \leq f\}$$

Equivalent to  $\max_\alpha \alpha$  s.t.  $h_{Y_\alpha}(F_i) \leq f_i \forall i \in \{1 \dots n\}$

**Theorem 6** (Geometric condition for invariance). Set  $\mathcal{O}$  is positively invariant set iff  $\mathcal{O} \subseteq \text{pre}(\mathcal{O}) \Leftrightarrow \text{pre}(\mathcal{O}) \cap \mathcal{O} = \mathcal{O}$

*Proof.* **Necessary** if  $\mathcal{O} \not\subseteq \text{pre}(\mathcal{O})$ , then  $\exists \bar{x} \in \mathcal{O}$  s.t  $\bar{x} \notin \text{pre}(\mathcal{O}) \rightsquigarrow \bar{x} \in \mathcal{O}, \bar{x} \notin \text{pre}(\mathcal{O})$ , thus  $\mathcal{O}$  not positively invariant

**Sufficient** if  $\mathcal{O}$  not positive invariant set, then  $\exists \bar{x} \in \mathcal{O}$  s.t  $g(\bar{x}) \notin \mathcal{O} \rightsquigarrow \bar{x} \in \mathcal{O}, \bar{x} \notin \text{pre}(\mathcal{O})$  thus  $\mathcal{O} \not\subseteq \text{pre}(\mathcal{O})$   $\square$

### 9.1 Control Invariance

**Definition 4** (Control Invariant Set).  $\mathcal{C} \subseteq \mathcal{X}$  control invariant if

$$x(k) \in \mathcal{C} \Rightarrow \exists u(k) \in \mathcal{U} \text{ s.t. } g(x(k), u(k)) \in \mathcal{C} \forall k$$

**Definition 5** (Maximal Control Invariant Set  $\mathcal{C}_\infty$ ). A set that contains all  $\mathcal{C}$  is the maximal positively invariant set  $\mathcal{C}_\infty \subset \mathcal{X}$

**Intuition** For all states in  $\mathcal{C}_\infty$  exists control law s.t constraints are never violated  $\rightsquigarrow$  **The best any controller could ever do**

**Pre-set**  $\text{pre}(S) := \{x \mid \exists u \in \mathcal{U} \text{ s.t. } g(x, u) \in S\}$

Set  $\mathcal{C}$  is control invariant iff:  $\mathcal{C} \subseteq \text{pre}(\mathcal{C}) \Leftrightarrow \text{pre}(\mathcal{C}) \cap \mathcal{C} = \mathcal{C}$

## Control Law from Control Invariant Set

Control law  $\kappa(x(k))$  will **guarantee** that the system with control invariant set  $\mathcal{C}$  satisfies constraints **for all time** if

$$x(k+1) = g(x(k), u(k)) \rightarrow g(x, \kappa(x)) \in \mathcal{C} \forall x \in \mathcal{C}$$

We can use this fact to **synthesize** control law  $\kappa$

$$\kappa(x) := \operatorname{argmin}\{f(x, u) \mid g(x, u) \in \mathcal{C}\}$$

with  $f$  as any function (including  $f(x, u) = 0$ )

Does not ensure that system will converge  
Difficult because calculating control invariant sets is hard  
**MPC** implicitly describes  $\mathcal{C}$  s.t easy to represent/compute

### 9.2 Robust Invariance

**Definition 6** (Robust Positive Invariant Set  $\mathcal{O}^\mathcal{W}$ ). For the autonomous system  $x(k+1) = f(x(k), w(k))$ , the set  $\mathcal{O}^\mathcal{W}$  is robust positive invariant if:

$$x \in \mathcal{O}^\mathcal{W} \Rightarrow f(x, w) \in \mathcal{O}^\mathcal{W}, \quad \forall w \in \mathcal{W}$$

Given set  $\Omega$  and dynamic system  $x(k+1) = f(x(k), w(k))$ ,

$$\operatorname{pre}^\mathcal{W}(\Omega) := \{x \mid f(x, w) \in \Omega \forall w \in \mathcal{W}\}$$

**Definition 7** (Robust Pre-Sets). The set of states that in the dynamic system  $x(k+1) = g(x(k), w(k))$  for all disturbance  $w \in \mathcal{W}$  in one time step evolves into the target set  $\Omega$  is the **pre-set** of  $\Omega \Rightarrow \operatorname{pre}^\mathcal{W}(\Omega) := \{x \mid g(x, w) \in \Omega \forall w \in \mathcal{W}\}$

**Theorem 7** (Geometric condition for robust invariance). Set  $\mathcal{O}^\mathcal{W}$  is robust positive invariant iff  $\mathcal{O}^\mathcal{W} \subseteq \operatorname{pre}^\mathcal{W}(\mathcal{O}^\mathcal{W})$

**Minimum Robust Invariant Set**

$$\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A_K^j \mathcal{W}, \mathcal{F}_0 := \{0\} \Rightarrow \mathcal{F}_n = \mathcal{F}_{n+1} = \mathcal{F}_\infty$$

### 9.3 Computing Invariant Sets and Pre-sets

**System for** **Pre-Set Computation**

$$x(k+1) = Ax(k) + Bu(k)$$

$$u(k) \in \mathcal{U} := \{u \mid Gu \leq g\}$$

$$S := \{x \mid Fx \leq f\}$$

**Invariant Pre-Set**

$$\begin{aligned} \operatorname{pre}(S) &:= \{x \mid Ax \in S\} \\ &= \{x \mid FAx \leq f\} \end{aligned}$$

**Control Invariant Pre-Set**

$$\begin{aligned} \operatorname{pre}(S) &:= \{x \mid \exists u \in \mathcal{U}, Ax + Bu \in S\} \\ &= \{x \mid \exists u \in \mathcal{U}, FAx + FBu \leq f\} \\ &= \left\{x \mid \exists u \in \mathcal{U}, \begin{bmatrix} FA & FB \\ 0 & G \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} \leq \begin{bmatrix} f \\ g \end{bmatrix}\right\} \end{aligned}$$

This is a **projection** operation

**System for** **Robust Pre-Set Computation**

$$\begin{aligned} x(k+1) &= Ax(k) + w(k) \\ \Omega &:= \{x \mid Fx \leq f\} \end{aligned}$$

### Robust Invariant Pre-Set

$$\begin{aligned} \operatorname{pre}^\mathcal{W}(\Omega) &= \{x \mid FAx + Fw \leq f\} \\ &= \{x \mid FAx \leq f - \max_{w \in \mathcal{W}} Fw\} \\ &= \{x \mid FAx \leq f - h_{\mathcal{W}^i}(F)\} \end{aligned}$$

where  $h_{\mathcal{W}^i}(F)$  is the **support function**  $\sigma_{\mathcal{C}}(a) = \sup_{x \in \mathcal{C}} a^\top x$

## Lyapunov

**Stability** is a property of an **equilibrium point**  $\bar{x}$  of a system

**Definition 8** (Lyapunov Stability).  $\bar{x}$  is **Lyapunov stable** if:  $\forall \epsilon > 0 \exists \delta(\epsilon)$  s.t.  $|x(0) - \bar{x}|_2 < \delta(\epsilon) \rightarrow |x(k) - \bar{x}|_2 < \epsilon$

**Definition 9** (Globally asymptotic stability). If  $\bar{x}$  is attractive, i.e.,  $\lim_{k \rightarrow \infty} \|x(k) - \bar{x}\| = 0, \forall x(0)$  and Lyapunov stable then  $\bar{x}$  is **globally asymptotically stable**.

**Definition 10** (Global Lyapunov function). For the equilibrium  $\bar{x} = 0$  of a system  $x(k+1) = g(x(k))$ , a function  $V$ , continuous at the origin, finite and such that  $\forall x \in \mathbb{R}^n$ :

$$|x| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty$$

$$V(x) = 0 \text{ if } x = 0 \quad \text{else} \quad V(x) > 0$$

$$V(g(x)) - V(x) \leq -\alpha(x)$$

for continuous positive definite  $\alpha: \mathbb{R}^n \rightarrow \mathbb{R}$

then  $V: \mathbb{R}^n \rightarrow \mathbb{R}$  is called **Lyapunov function**.

**Theorem 8.** If a system admits a Lyapunov function  $V(x)$ , then  $\bar{x} = 0$  is **globally asymptotically stable**.

**Theorem 9** (Lyapunov indirect method). System linearized around  $\bar{x} = 0$  with resulting matrix  $A$  and eigenvalues  $\lambda_i$ . If  $\forall |\lambda_i| < 1$  then the origin is asymptotically stable. if  $\exists |\lambda_i| > 1$  then origin is unstable. If  $\exists |\lambda_i| = 1$  we can't conclude anything about stability.

**Discrete-Time Lyapunov equation**  $A^\top PA - P = -Q, \quad Q > 0 \quad (3)$

**Theorem 10** (Existence of solution, DT Lyapunov equation). The discrete-time Lyapunov equation has a unique solution  $P > 0$  iff the system  $x(k+1) = Ax(k)$  is stable.

## 10 Optimization

### Mathematical Optimization Problem

**Decision variable**  $x \in \mathbb{R}^n$   
**Objective function**  $f: \operatorname{dom}(f) \rightarrow \mathbb{R}$   
**Inequality constraints**  $g_i \ (i \in \# \text{constraints})$   
**Equality constraints**  $h_i \ (i \in \# \text{constraints})$   
**Feasible set**  $\mathcal{X} := \{x \mid g(x) \leq 0, h(x) = 0\}$

**minimize**  $f(x)$   
**subject to:**  
 $g_i(x) \leq 0$   
 $h_i(x) = 0$

**Feasible point**  $x \in \operatorname{dom}(f)$  with  $g_i(x) \leq 0, h_i(x) = 0$   
**Strictly feasible point**  $x$  with strict inequality  $g_i(x) < 0$   
**Optimal value**  $f^*$  (or  $p^*$ )  $= \inf\{f(x) \mid g_i(x) \leq 0, h_j = 0\}$   
 $f^* = +\infty$ : OP infeasible,  $f^* = -\infty$ : OP unbound below  
**Optimizer set:**  $\operatorname{argmin}_{x \in \mathcal{X}} f(x) := \{x \in \mathcal{X} \mid f(x) = f^*\}$

$x^*$  is a **Global Minimum** if  $f(x^*) \leq f(x)$   
 $x^*$  is a **Local Minimum** if  $\exists \epsilon > 0$  s.t.  $f(x^*) \leq f(x)$   
 $\forall x \in \mathcal{X} \cap B_\epsilon(x^*)$ , open ball with center  $x^*$  and radius  $\epsilon$

### 10.1 Convex Sets

**Definition 11** (Convex Set). Set  $\mathcal{C}$  is convex if and only if

$$\theta x + (1 - \theta)y \in \mathcal{C}, \quad \forall x, y \in \mathcal{C}, \quad \forall \theta \in [0, 1]$$

**Intersection**  $\mathcal{C}_1, \mathcal{C}_2$  cv  $\Rightarrow \mathcal{C}_1 \cap \mathcal{C}_2$  convex (**cv**)

**Image under affine map**  $\mathcal{C} \subseteq \mathbb{R}^n$  cv  $\Rightarrow \{Ax + b \mid x \in \mathcal{C}\}$  cv

**Inverse loaM**  $\mathcal{C} \subseteq \mathbb{R}^n$  cv  $\Rightarrow \{x \in \mathbb{R}^n \mid Ax + b \in \mathcal{C}\}$  cv

**Definition 12** (Hyperplanes).  $\{x \in \mathbb{R}^n \mid a^\top x = b\}$

**Definition 13** (Halfspaces).  $\{x \in \mathbb{R}^n \mid a^\top x \leq b\}$  can be **open** (strict inequality) or **closed** (non-strict inequality)

**Definition 14** (Polyhedra). intersection of **finite** number of closed halfspaces: polyhedra  $\{x \in \mathbb{R}^n \mid A^q \times n x \leq b^{q \times 1}\}$

**Definition 15** (Polytope). is a **bounded** polyhedron.

**Definition 16** (Convex hull). for  $\{v_1, \dots, v_k\} \in \mathbb{R}^d$  is:

$$\operatorname{co}(\{v_1, \dots, v_k\}) := \{x \mid x = \sum_i \lambda_i v_i, \lambda \geq 0, \sum_i \lambda_i = 1\}$$

**Definition 17** (Ellipsoid). set:  $\{x \mid (x - x_c)^\top A^{-1} (x - x_c) \leq 1\}$  where  $x_c$  is center of ellipsoid,  $A \succ 0$  (i.e. positive definite)

(Semi-axis lengths are square roots of eigenvalues of  $A$ )

**Definition 18** (Norm Ball).  $B_r(x) := \{\xi \in \mathbb{R}^n \mid \|\xi - x\|_p < r\}$  where  $p$  defines the  $l_p$  norm,  $p = \{1, 2, \dots, \infty\}$

**Theorem 11.** Minkowski-Weyl

The following statements are equivalent for  $\mathcal{P} \subseteq \mathbb{R}^d$

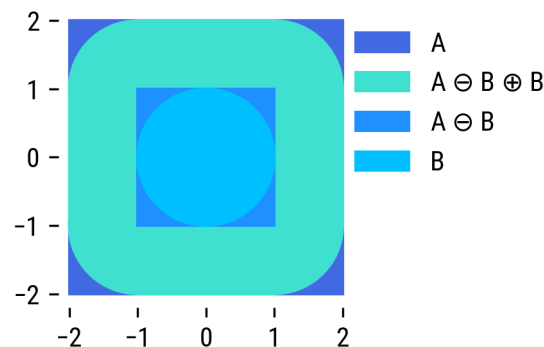
$\mathcal{P}$  is a polytope and there exists  $A, b$  s.t  $\mathcal{P} = \{x \mid Ax \leq b\}$

$\mathcal{P}$  finitely generated,  $\exists$  finite set  $\{v_i\}$  s.t  $\mathcal{P} = \operatorname{co}(\{v_1, \dots, v_s\})$

**Definition 19.** **Minkowski Sum** for  $A, B \subset \mathbb{R}^n$

$$A \oplus B := \{x + y \mid x \in A, y \in B\}$$

**Example**  $[a, b] \oplus [c, d] = [a + c, b + d]$



**Definition 20.** **Pontryagin Difference** for  $A, B \subset \mathbb{R}^n$

$$A \ominus B := \{x \mid x + e \in A, \forall e \in B\}$$

**Example**  $[a, b] \ominus [c, d] = [a - c, b - d]$



10.2 Convex Functions

**Definition 21** (Convex Function).  $f : \mathcal{C}_{\text{convex}} \rightarrow \mathbb{R}$  is convex iff  $f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y), \forall x, y \forall \theta \in [0, 1]$   
 $f$  is strictly convex if this inequality is strict.  
**Definition 22** (Epigraph).  $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \infty \Leftrightarrow \text{epi}(f)$  is cv set  
 $\text{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} | f(x) \leq t\}$

**Check Convexity**  $f$  is convex if it is composition of simple convex function with convexity preserving operations or if  
 $f : \mathbb{R}^n \rightarrow \mathbb{R}$  twice differentiable,  $\partial^2 f / \partial x^2 \succeq 0 \forall x \in \mathbb{R}^n$   
 $g : \mathbb{R} \rightarrow \mathbb{R}$  with  $g(t) = f(x + tv)$  convex in  $t \forall x, v \in \mathbb{R}^n \rightarrow f$  convex (restriction to a line)  
- the point wise maximum of convex functions is convex  
- the sum of convex functions is convex  
-  $f(Ax + b)$  is convex if  $f$  is convex

10.3 Optimality Conditions

Lagrange Duality

Consider  $f^* = \inf_{x \in \mathbb{R}^n} f(x)$  s.t.  $g(x) \leq 0, h(x) = 0$  (4)

Lagrangian  $\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T h(x)$

Dual Function  $d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu)$

**Proposition 1** (Weak Duality).  $d(\lambda, \nu) \leq f^*, \forall \lambda \geq 0, \nu \in \mathbb{R}^h$   
**Definition 23** (Constraint qualification). **Slater's Condition** holds if  $\exists$  at least one **strictly feasible point**  $\hat{x}$  ( $h(\hat{x}) = 0, g(\hat{x}) < 0$ )  
**Proposition 2** (Strong Duality). If Slater's condition holds and OP is convex  $\Rightarrow \exists \lambda \geq 0, \nu \in \mathbb{R}^{n_h}$  s.t.  $d(\lambda, \nu) = f^*$

KKT Conditions (Karush-Kuhn-Tucker)

Theorem 12 (KKT Conditions). If Slater's condition holds and (4) is convex  $\rightarrow x^* \in \mathbb{R}^n$  is a minimizer of the primal (4) and  $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^{n_g} \times \mathbb{R}^{n_h}$  is a maximizer of the dual  $\Leftrightarrow$  is equivalent to the following statements:

KKT-1 (Stationary Lagrangian)  $\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$   
KKT-2 (primal feasibility)  $g(x^*) \leq 0, h(x^*) = 0$   
KKT-3 (dual feasibility)  $\lambda^*, \nu^* \in \mathbb{R}^{n_h} \geq 0$   
KKT-4 (complementary slackness)  $\lambda^{*T} g(x^*) = 0, \nu^{*T} h(x^*) = 0$

In addition we have:  $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^{n_h}} q(\lambda, \nu) = \inf_{x \in \mathcal{C}} f(x)$

**Remark** Without Slater, KKT1-4 still implies  $x^*$  minimizes (4) and  $\lambda, \nu$  maximizes dual, but the converse is no longer true. There can be primal-minimizer/dual-maximizer not satisfy KKT.

10.4 Convex Optimization Problems

**Theorem 13.** For a convex optimization problem, **any** locally optimal solution is globally optimal (local optima are global optima).

**Linear Programming** minimize  $c^T x$  s.t.  $Ax - b \geq 0, x \geq 0$   
Step 1:  $\mathcal{L}(x, \lambda_1, \lambda_2) = c^T x - \lambda_1^T (Ax - b) - \lambda_2^T x, \lambda_i \geq 0$

Step 2:  $\inf_{x \in \mathbb{R}^n} \mathcal{L} = \lambda_1^T b$ , if  $c - A^T \lambda_1 - \lambda_2 = 0$ , else  $-\infty$   
Step 3: Dual, maximize  $b^T \lambda$  s.t.  $c - A^T \lambda \geq 0, \lambda \geq 0$  (again LP)  
**Proposition 3.** The optimal solution of a linear program (if it exists) lies always on the boundary of the feasible set and there exists an optimal solution that is a vertex of the feasible set.

**Quadratic Programming** convex if  $P = P^T$  positive semi-definite minimize  $\frac{1}{2} x^T P x + q^T x$  s.t.  $Gx \leq h, Ax = b$   
**Second-Order Cone Program**  
minimize  $f^T x$  s.t.  $|A_i x + b| \leq c_i^T x + d_i, Fx = g$   
Second-order cone  $C_{n+1} = \{(x, t) | x \in \mathbb{R}^n, t \in \mathbb{R}, |x| \leq t\}$   
 $|A_i x + b| \leq c_i^T x + d_i \Leftrightarrow (A_i x + b, c_i^T x + d_i) \in C_{n+1}$   
**Semi-Definite Programming** with symmetric  $F_i, X, A_i$   
minimize  $c^T x$  s.t.  $\sum_{i=1}^n x_i F_i + G \preceq 0, Ax = b$   
**CFTOC Optimization**  
**CFTOC** problem is **multiparametric quadratic program mp-QP**

Two options to solve it:  
- Iterative optimization methods  
- Explicit solution  
Let  $I := 1, \dots, m$  be the set of constraint indices.  
**Definition 24** (Active Set).  $A(x)$  and its complement  $NA(x)$   
 $A(x) := \{j \in I : G_j z^*(x) - S_j x = w_j\}$   
 $NA(x) := \{j \in I : G_j z^*(x) - S_j x < w_j\}$

**Definition 25** (Critical Region).  $CR_A$  is set of parameters  $x$  for which set  $A \subseteq I$  of constraints  $i$  active at the optimum. For given  $\bar{x} \in \mathcal{K}^*$  let  $(A, NA) := (A(\bar{x}), NA(\bar{X}))$ . Then  
 $CR_A := \{x \in \mathcal{K}^* : A(x) = A\}$  (states share active set)

**Online evaluation: Point location**  
Sequential search  
Logarithmic search  
**Large Scale Convex Optimization**  
**Definition 26** (Lipschitz continuity).  $q : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is Lipschitz with constant  $L$  if:  $|q(x) - q(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$

$f$  is **Lipschitz** (Lip) with constant  $L \Leftrightarrow |\nabla f(x)|_2 \leq L$   
**Theorem 14.**  $\mathcal{C} \subseteq \mathbb{R}^n$  non-empty closed (cl) convex set,  $y \notin \mathcal{C} \rightarrow \exists a \neq 0, b \in \mathbb{R}$  s.t.  $a^T x + b < a^T y + b, \forall x \in \mathcal{C}$   
**Definition 27** (smoothness).  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is  $L$ -smooth ( $L$ -sm) if  $\nabla f(x)$  satisfies  $|\nabla f(x) - \nabla f(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$   
Taylor  $\rightarrow f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} |x - y|^2$   
**Definition 28** (strong convexity).  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is  $\mu$ -strongly convex ( $\mu$ -scv) if  $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} |x - y|^2$

**How to find  $\mu/L$ , Spectra of Hessian  $\nabla^2 f$ , min/max eigenvalue**  
**Gradient Descent**  
 $x_{k+1} = x_k - T \nabla f(x_k)$  for  $k = (k_0, \dots, k_N)$  given  $x_0, T$   
**Line search** optimal step  $\nu_i^* = \text{argmin}_{\nu \in \mathbb{R}} f(x_t - \nu \nabla f(x_t))$   
**Definition 29.**  $\text{prox}_{\mathcal{C}}(x) = \text{argmin}_{y \in \mathcal{C}} \frac{1}{2} |x - y|^2, \mathcal{C} \subset \mathbb{R}^n$   
**Lemma 3.** cl, cv  $\mathcal{C} \subset \mathbb{R}^n \rightarrow |\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y)| \leq |x - y| \leftarrow |\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y)|^2 \leq (\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y))^T (x - y)$   
**Projected Gradient Descent**  
 $x_{k+1} = \text{prox}_{\mathcal{C}}(x_k - T \nabla f(x_k)),$  for  $x_0, k_{0..N}, T \in (0, 2/L)$

11 Introduction to Systems and Controls

**Idea** Create a model by solving the systems physical equations  
$$x(t) = e^{A^c(t-t_0)} x_0 + \int_{t_0}^t e^{A^c(t-\tau)} B^c u(\tau) d\tau$$
  
(Exact Solution to ODE of a Linear System)  
**Problem** Most physical systems are nonlinear  
**Trick** use First Order Taylor expansion  $f(\bar{x}) + \frac{\partial f}{\partial x^T} \Big|_{\bar{x}} (x - \bar{x})$

**11.1 Linearization**  
**Idea** Nonlinear system stable enough around an equilibrium  
System equations  $\dot{x}_s = g(x_s, u_s) = 0, y_s = h(x_s, u_s)$   
Find stationary operating point  $x_s, u_s$  and plug in derivative:

$$\begin{aligned} \Delta \dot{x} &= \dot{x} - \dot{x}_s & A^c &= \frac{\partial g}{\partial x^T} \Big|_{x_s} & B^c &= \frac{\partial g}{\partial u^T} \Big|_{x_s} \\ &= A^c \Delta x + B^c \Delta u & C &= \frac{\partial h}{\partial x^T} \Big|_{x_s} & D &= \frac{\partial h}{\partial u^T} \Big|_{x_s} \\ \Delta y &= y - y_s \end{aligned}$$

**11.2 Discretization**  
For general nonlinear systems only approximate discretization methods exist, such as Euler, quality depends on sampling time

**Approximation**  $\dot{x}^c \approx \frac{x^c(t + T_s) - x^c(t)}{T_s}$  **Notation**  $x(k) := x^c(t_0 + kT_s)$   
 $u(k) := u^c(t_0 + kT_s)$

**Exact Discretization of Linear Time-Invariant Models**  
$$x(t_{k+1}) = \underbrace{e^{A^c T_s}}_{=A} x(t_k) + \underbrace{\int_0^{T_s} e^{A^c(T_s-\tau)} B^c d\tau}_{B=(A^c)^{-1}(A-I)B^c} u(t_k)$$
  
$$x(k + N) = A^N x(k) + \sum_{i=0}^{N-1} A^i B u(k + N - 1 - i)$$

**11.3 Analysis of Discrete-Time LTI Systems**  
**Controllable** if  $\text{rank}(\mathcal{C}) = n, \mathcal{C} = [B \dots A^{n-1}B]$   
 $\forall (x(0), x^*) \exists$  finite time  $N$  with inputs  $\mathcal{U}$ , s.t.  $x(N) = x^*$   
**Stabilizable** iff all uncontrollable modes stable  
**Observable** if  $\text{rank}(\mathcal{O}) = n, [C^T \dots (CA^{n-1})^T]^T$   
 $\forall x(0) \exists$  finite time  $N$ , s.t. the measurements  $y(0), \dots, y(N-1)$  uniquely distinguish initial state  $x(0)$   
**Detectable** iff all unobservable modes stable

12 Optimal Control

Discrete-Time Optimal Control Problem

Cost Function  $J(x_0, U) = \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N)$  (5)

Stage Cost  $l_f(x_i, u_i)$

Terminal Cost  $l_f(x_N)$

Constraints  $x_{i+1} = g(x_i, u_i)$   
 $x_0 = x(k)$   
 $h(x_i, u_i) \leq 0$  (optional)

12.1 Unconstrained Finite Horizon Control Problem

Linear Quadratic Optimal Control

Cost Function  $J^*(x(0)) := \min_U \sum_{i=0}^{N-1} x_i^T Q x_i + u_i^T R u_i + x_N^T P x_N$  (6)

Terminal weight  $P \succeq 0$  symmetric

State weight  $Q \succeq 0$  symmetric

Input weight  $R \succ 0$  symmetric

Constraints No input or state constraints!  
 $x(k+1) = Ax_k + Bu_k$   
Only dynamics matter.

**Batch Approach**  
express cost function in terms of  $x(0)$  and input sequence  $U$

$$\begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} I \\ A \\ \vdots \\ A^N \end{bmatrix} x(0) + \begin{bmatrix} 0 & \dots & 0 \\ B & 0 & 0 \\ AB & B & 0 \\ \vdots & \ddots & 0 \\ A^{N-1}B & \dots & B \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \vdots \\ u_{N-1} \end{bmatrix}$$

$X := S^x x(0) + S^u U \quad J(x(0), U) = X^T \bar{Q} X + U^T \bar{R} U$   
 $\bar{Q} := \text{blockdiag}(Q, \dots, Q, P) \quad \bar{R} := \text{blockdiag}(R, \dots, R)$   
**Optimal Input** set  $\nabla_U J(x(0), U) = 2HU + 2F^T x(0) = 0$   
$$U^*(x(0)) = - \underbrace{((S^u)^T \bar{Q} S^u + \bar{R})^{-1}}_{H(\text{Hessian})^{-1}} \underbrace{(S^u)^T \bar{Q} S^x}_{F^T} x(0)$$

**Optimal Cost** ( $x_0 = x(0)$ )  
 $J^*(x_0) = x_0^T (S_x^T \bar{Q} S_x - S_x^T \bar{Q} S_u (S_u^T \bar{Q} S_u + \bar{R})^{-1} S_u^T \bar{Q} S_x) x_0$

Ricatti Equations

Riccati Difference Equation - RDE solved recursively  
 $P_i = A^T P_{i+1} A + Q - A^T P_{i+1} B (B^T P_{i+1} B + R)^{-1} B^T P_{i+1} A$

Algebraic Riccati Equation - ARE solved analytically  
 $P_\infty = A^T P_\infty A + Q - A^T P_\infty B (B^T P_\infty B + R)^{-1} B^T P_\infty A$

**Recursive Approach**  
use dynamic programming to solve problem backwards from  $N$

$$J_j^*(x(j)) := \min_{U_{j \rightarrow N}} x_N^T P x_N + \sum_{i=j}^{N-1} x_i^T Q x_i + u_i^T R u_i$$

**From Principle Of Optimality** **Optimal Cost-To-Go**  
 $J_j^*(x_j) = \min_{u_j} I(x_i, u_i) + J_{j+1}^*(x_{j+1}) \quad J_i^*(x_i) = x_i^T P_i x_i$   
**Optimal Control Policy** use Riccati  
$$u_i^* = F_i x_i = -(B^T P_{i+1} B + R)^{-1} B^T P_{i+1} A \cdot x(i)$$

**Comparison of Batch and Recursive Approach**  
Dynamic programming yields **feedback policies**  $u_i^* = F_i x_i$  depending on each  $x_i$ . Batch optimization returns sequence  $U^*(x(0))$  of **numeric values** depending only on  $x(0)$ ,  
**12.2 Infinite Horizon Control Problem**

## Linear Quadratic Regulator

### Cost Function

$$J_{\infty}^*(x(k)) = \min_{u(\cdot)} \sum_{i=0}^{\infty} x_i^{\top} Q x_i + u_i^{\top} R u_i \quad (7)$$

$$\text{Constraints} \quad x_{i+1} = A x_i + B u_i \\ x_0 = x(k)$$

$$\text{Optimal Input} \quad u^*(k) = F_{\infty} x(k) \quad (\text{use Riccatti})$$

$$F_{\infty} x(k) = -(B^{\top} P_{\infty} B + R)^{-1} B^{\top} P_{\infty} A x(k)$$

**Lemma 4** (Lyapunov function for LQR). If  $(A, B)$  is stabilizable and  $Q, R \succ 0$  then  $J^*(x) = x^{\top} P_{\infty} x$  is a **Lyapunov function** for the system  $x^+ = (A + B F_{\infty})x$  where  $F_{\infty}$  is the constant feedback matrix and  $P_{\infty} \succ 0$  solves the Riccatti equation..

$$\begin{aligned} \mathcal{X} &= \{x \mid A_x x \leq \mathcal{X}_f q b_x\} \\ \mathcal{U} &= \{u \mid A_u u \leq b_u\} \\ \mathcal{X}_f &= \{x \mid A_f x \leq b_f\} \end{aligned} \quad E_{in} = \begin{bmatrix} -A_x \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$G_{in} = \left[ \begin{array}{cc|cc} 0 & & 0 & \\ A_x & & 0 & \\ & \ddots & & \\ & & A_x & \\ \hline 0 & & A_u & 0 \\ & \ddots & & \\ & & 0 & 0 \end{array} \right] w_{in} = \begin{bmatrix} b_x \\ b_x \\ \vdots \\ b_x \\ b_f \\ b_u \\ \vdots \\ b_u \end{bmatrix}$$

$$\text{Cost Matrix } \bar{H} = \text{diag}(Q, \dots, Q, P, R, \dots, R)$$

Finally the resulting quadratic optimization problem

$$\begin{aligned} J^*(x(k)) &= \min_z \begin{bmatrix} z^{\top} & x(k)^{\top} \end{bmatrix} \begin{bmatrix} \bar{H} & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} z^{\top} & x(k)^{\top} \end{bmatrix}^{\top} \\ \text{s.t. } G_{in} z &\leq w_{in} + E_{in} x(k) \quad G_{eq} z = E_{eq} x(k) \end{aligned}$$

**Construction of QP with substitution**

**Idea** Substitute the state equations.

**Step 1** Rewrite cost as

$$\begin{aligned} J(x(k)) &= U^{\top} H U + 2x(k)^{\top} F U + x(k)^{\top} Y x(k) \\ &= \begin{bmatrix} U^{\top} & x(k)^{\top} \end{bmatrix} \begin{bmatrix} H & F^{\top} \\ F & Y \end{bmatrix} \begin{bmatrix} U^{\top} & x(k)^{\top} \end{bmatrix}^{\top} \end{aligned}$$

**Step 2** Rewrite constraints compactly as  $GU \leq w + Ex(k)$

$$G = \begin{bmatrix} A_u & 0 & \dots & 0 \\ 0 & A_u & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & A_u \\ 0 & 0 & \dots & 0 \\ A_x B & 0 & \dots & 0 \\ A_x A B & A_x B & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ A_f A^{N-1} B & A_f A^{N-2} B & \dots & A_f B \end{bmatrix}, E = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ -A_x \\ -A_x A \\ -A_x A^2 \\ \vdots \\ -A_f A^N \end{bmatrix}, w = \begin{bmatrix} b_u \\ b_u \\ \vdots \\ b_u \\ b_x \\ b_x \\ b_x \\ \vdots \\ b_f \end{bmatrix}$$

**Step 3** Rewrite constrained problem as

$$\begin{aligned} J^*(x(k)) &= \min_U \begin{bmatrix} U^{\top} & x(k)^{\top} \end{bmatrix} \begin{bmatrix} H & F^{\top} \\ F & Y \end{bmatrix} \begin{bmatrix} U^{\top} & x(k)^{\top} \end{bmatrix}^{\top} \\ \text{subj. to } GU &\leq w + Ex(k) \end{aligned}$$

## 12.3 Constrained Finite Time Optimal Control Problem

### Choice of P

- 1 Match the infinite solution  $P_N = P_{\infty}$
- 2 Use solution of the Lyapunov Equation (3) if the system is asymptotically stable (otherwise P not positive definite), assumes no control needed after end of horizon.
- 3 Set  $P = 0$  and use instead constraint  $x_{i+N} = 0$

### Quadratic Cost CFTOC

$$J(x(k)) = x_N^{\top} P x_N + \sum_{i=0}^{N-1} x_i^{\top} Q x_i + u_i^{\top} R u_i$$

### CFTOC

**Cost Function** equal to Linear Quadratic Optimal Control (6)  
**Constraints** as in (6)  $\oplus x_N \in \mathcal{X}_f, x_{i < N} \in \mathcal{X}, u_i \in \mathcal{U}$

**Goal: Transform into QP**

$$\min_{z \in \mathbb{R}^n} \frac{1}{2} z^{\top} H z + q^{\top} z + r \quad \text{s.t. } G z \leq h, A z = b$$

**Construction of QP without Substitution**

**Idea** Keep state equations as equality constraints

$$\text{Define variable } z = \begin{bmatrix} x_1^{\top} & \dots & x_N^{\top} & u_0^{\top} & \dots & u_{N-1}^{\top} \end{bmatrix}^{\top}$$

**Equalities** from system dynamics  $x_{i+1} = A x_i + B u_i$

$$G_{eq} = \left[ \begin{array}{c|c} \begin{matrix} \mathbb{I} & & \\ -A & \mathbb{I} & \\ & \ddots & \ddots \\ & & -A & \mathbb{I} \end{matrix} & \begin{matrix} -B \\ & \ddots & -B \end{matrix} \end{array} \right] E_{eq} = \begin{bmatrix} A \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

**Inequalities**  $G_{in} z \leq w_{in} + E_{in} x(k)$  for  $\mathcal{X}, \mathcal{U}$